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Covid Econometrics

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Working Paper

"THE EFFECT OF WEARING MASK TO PROTECT THEMSELVES FROM THE COVID-19 INFECTION ON THE 7-DAY INCIDENCE RATE: EVIDENCE FROM ARKANSAS STATE"

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ABSTRACT

As a response to the Covid-19 crisis, countries around the world started implementing a series of Non-Pharmaceutical Interventions (NPIs). Measures such as social distancing, travel restrictions, lockdowns, schools and universities closure in order to mitigate the transmission of the virus.

Even though it was hard at the beginning to say whether or not face masks requirements appear to be effective, quantitative research is still needed to assess the strategy of NPIs and their timing.

In this term project for Covid Econometrics class we have analyzed whether a state-wide face covering mandate has an effect on the 7-day incidence rate.

We have focused on the Executive Order 20-43 signed by the Governor of Arkansas, requiring a state-wide face mask requirement from the 20th of July 2020 and we have developed our analysis on how mask wearing has affected number of new daily infections from Covid-19.

We used two different methods: the Difference-in-Differences design and the Synthetic Control Group method.

In the Difference-in-Differences approach, we have run three different specifications using three different control states, Missouri, Tennessee and Oklahoma. We selected those three states since none of them has issued a similar treatment during that period; moreover, according to a recent study, those three states are the most similar to Arkansas in all United States.

The DiD results using Tennessee and then Oklahoma as control group were not significant, while using Missouri as control, the DiD design shows a significant variation in the 7 days incidence rate, with a decrease of almost 30 new cases in the 7 days incidence rate every 100,000 inhabitants.

Then, we have developed the Synthetic Control Group method. The first specification included also some states which have issued during that period of time the same treatment: since this represents a violation of the common trend assumption, we have kept only the states that could have been a good control. The second specification showed an increase in the 7-day incidence rate and a later decrease after twenty days from the intervention day.

In order to providing support for the identification assumption in the Synthetic Control Group method, we have conducted two placebo regressions: in-time placebo test, where the intervention artificially takes place one month earlier, and the in-space placebo test, where the intervention artificially takes place in a control group from the donor pool. In both cases, the

confidence that the face covering intervention has an effect on the outcome is rejected since both graphs the curves deviates, so we cannot state that the change is due to the treatment itself.

INTRODUCTION

From the very start, as the coronavirus reached the United States, America demonstrated a very hard epidemiological situation. The situation in some of the states was sometimes worse than the average situation in some other countries. The United States is just 4% of the world population but had a quarter of its confirmed cases and deaths (Yong, 2020)
The number of new cases has skyrocketed due to several reasons. A slow start from the federal government, an inefficient health-care system that has been underfunded for years, misinformation and conspiracy theories via social media platforms are just to name a few. As the virus spread from state to state a combination of these factors played a crucial role in trying to contain the virus and saving lives of those who had to be hospitalized. The ongoing protests that intensified after police brutality against minorities and the presidential race for the upcoming elections later that year only showed more ineffectiveness of the federal government to respond to the spread of Covid-19 accordingly.

Since the start of Covid-19, on March 11th, the World Health Organization declared coronavirus a pandemic, governments throughout the world started implementing non-pharmaceutical interventions to contain the outbreak. The question they had to face was which one of these strategies based on non-pharmaceutical interventions could help to stop the spread of the disease since the effectiveness of different interventions varied (Lai et al., 2020) Implementing such policies could be achieved through different preventive measures.
One of the main challenges that politicians had to face, at the very beginning of the Covid-19 outbreak, was trying to save the economic activity while protecting vulnerable classes of the population at the same time. Social distancing and face covering with masks were one of the main tools in combating the virus from spreading at the beginning of the pandemic in 2020, along with physical distancing, use of hand sanitizer and avoiding touching surfaces especially in public settings.
Face coverings, even in the form of a homemade mask, are very important when it comes to preventing diseases that transmit via airborne pathogens. (Davies et al., 2013)

According to the Centers for Disease Control and Prevention even when America was recording thousands new cases at the early stages of the pandemic, asking people to wear face masks at public places was not as easy as one would expect. Wearing a face covering in public was a controversial and politicized issue in the United States, even with case evidence from other countries that face coverings help to control the spread of the coronavirus, writes Michael Haischer at Marquette University. (Haischer et al., 2020)

In this paper we want to test whether or not implementing face masks requirements prevents a high number of daily new cases by using econometrics tools as methods for evaluation of the intervention.

Specifically, we will argue that requiring wearing a facial coverings/face masks in the state of Arkansas has resulted in a lower average of 7-day incidence rate compared to the control states during the same period of time.

The paper is divided in four main parts. First, the literature review introduces to the reader the topic we have developed including the most relevant working papers published in the last two years about face covering mandates and their consequences. The paragraph about "Covid-19 situation in Arkansas" puts the attention on how the local Government in Arkansas faced the Covid-19 health emergency. The next section focuses on the empirical strategies and on the two methods used to test the research question: the Difference-in-Differences approach and the Synthetic Control Group method. This part also includes information about the data and variables used. In the last part, we present the main result.

LITERATURE REVIEW

When a person is infected, talking, sneezing and coughing produces droplets that can carry enough of the virus to infect others around them. The same way Covid-19 infection can be transmitted through two different mechanisms, both of which include spreading the droplets from an infected person to an uninfected one. It can be that the droplets of the virus are hanging in the air and therefore inhaled by others, or they fall and remain on the surface of objects invisibly. These surfaces can then be touched, usually with hands, by other people later before completely evaporating. If worn properly a face mask can effectively prevent droplets from leaving the mask and spreading around the mask-wearer. The face mask's effectiveness is

caused by the droplet size being large enough that they can easily be trapped by a face covering, however (Ng, 2021)

There is no clear evidence showing wearing a face mask alone would stop the virus from spreading quickly. Also, there is hardly any proof from laboratory studies on the success of masks as a treatment. Thus, the collected data from such studies sometimes might lead to different conclusions. Ideally, we would need randomized controlled trials where we could study people with and without masks.
Researchers at the Hong Kong University wanted to test exactly this but on hamsters instead. They divided healthy and infected ones into different cages with a face mask stretched in between. They would then start a fan blowing and see if any of the healthy animals got infected with Covid-19 (Chan et al., 2020)
Two-thirds did get infected within a week when a face mask was not in the air stream. And only one-third were infected when a face mask was in place (The Economist, 2020).
Anna Davies et al published an article in 2013 showing how the usage of homemade masks would protect in an influenza pandemic. A face covering is very important when it comes to preventing diseases that transmit via airborne pathogens (Davies et al., 2013). Dasgupta et al found that counties in states with statewide mask mandates had a 43% lower chance of having rapid case growth based on the data obtained from state government websites and by using Poisson regression models.

To reduce the spread of Covid-19, many health officials have recommended covering faces with masks. Implementing wearing a face mask as a strategy of non-pharmaceutical intervention is an essential component of the public health response to Covid-19 outbreaks (Flaxman et al., 2020) Evidence from a natural experiment on the effects of state government mandates for face masks by 15 states plus DC between April 8 and May 15, 2020 shows mask use in public is associated with a decline in the daily covid-19 growth rate by two percentage points in 21 or more days after state face mask orders were signed (Lyu and Wehby, 2020).
Wearing masks has previously met resistance and public mask wearing has been labeled as not effective (Sunjaya and Jenkins, 2020) Indeed, there are some other reasons and arguments that were used to support such negative opinions from the public. People's refusal to follow the advice from the government was also based on existing cultural frames assessed at both the individual level and the state level (Kemmelmeier and Jami, 2021). Marcus Painter and Tian Qiu in the Journal of Economic and Organization by using machine learning approaches

showed how political polarization can affect public policy responses and the correlation of refusal to wear a mask with partisanship. The study further investigated whether Trump's unexpected mask use at Walter Reed on July 11, 2020 and his later endorsing of masks had a positive effect on mask-related topics in the public.

COVID 19 SITUATION IN ARKANSAS

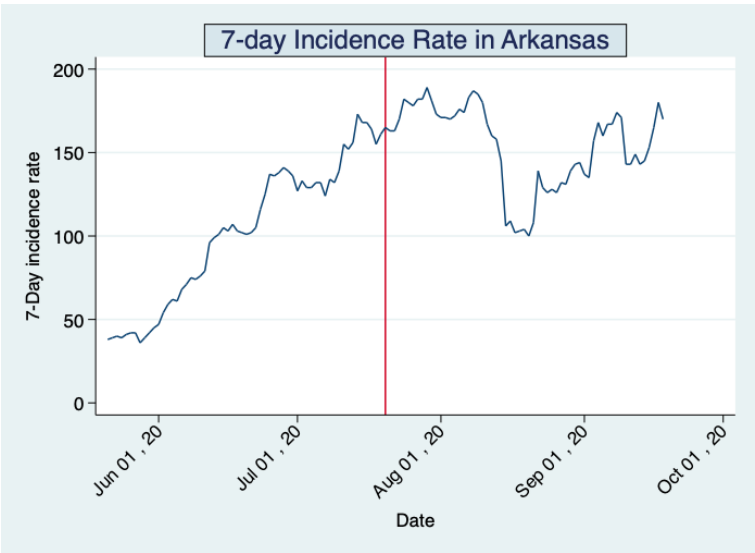
Having seen the federal government doing a poor job at containing the spread of the virus even though they have had all the tools and resources available to tackle the pandemic in a more efficient way, the local governments had to take control of the situation.
Weak coordination between national and state guidelines only added more fuel to an already poor Covid-19 handling, which left local authorities to face a trade-off between epidemiological costs (lives) and economic costs (livelihoods).
The federal government lacks the legal authority to impose strict lockdowns which leaves local states to decide on their own what measures to implement against the spread of the pandemic. Southern states like Arkansas faced an even bigger challenge with most of its population being self-employed.

The US Arkansas State ordered a state-wide mask requirement from 20th of July 2020 to 30th of March 2021 to mitigate the spread of Covid-19 as the number of new cases per day was rapidly increasing[1]. On the 20th March 2020, the Governor issued the closure of schools, dine-in activities in bars and restaurants, gyms and indoor entertainment venues. On the 14th of April, the Governor issued an Executive Order requiring businesses, manufacturers, construction companies and places of worship to implement social distancing protocols, such as limiting the number of people who may enter a facility at once and providing hand sanitizers near the entrance. From the end of April, the Governor announced a series of openings regarding restaurants, bars, gyms. The openings took place on the 15th June 2020. Since Covid-19 cases continued to spike, the executive power paused further reopening and started to permit cities to implement ordinances about face mask requirements by the beginning of July.
On the 16th of July, the Governor of Arkansas, Asa Hutchinson, signed the "Executive Order 20-43 pursuant to the public health emergency concerning Covid-19, as declared in Executive

[1] From HuschBlackwell website "COVID-19 State-by-State Daily Updates Archive", available from: https://www.huschblackwell.com/state-by-state-daily-updates-archive

Order 20-37, for the purpose of requiring face covering". According to this statement, "the Secretary of Health shall issue a public health directive requiring every person in Arkansas to wear a face covering over the mouth and nose in all indoor environments where they are exposed to non-household members and distancing of six feet or more cannot be assured and in all outdoor settings where there is exposure to non-household members, unless there exists ample space of six feet or more to practice physical distancing". The executive order also clarified the cases of exemption from the order.

The state's initial response to the pandemic was slow and imprecise. Even after governor Hutchinson signed an executive order to declare a public health emergency after the first positive case of the virus was confirmed in early March 2020, a lot more could have been done in order to be prepared for the spread of the virus. The state health department secretary Nate Smith said that restrictions may need to be in place longer than those in some other states, while the governor said that he would like to see life back to normal by Easter (April 12th) On that day when the national social distancing guidelines were released (March 16th) two cases were confirmed in a small county of 25,000 people in Arkansas. They were the very first confirmed cases of coronavirus in this county (James et al., 2020). It has been found a positive result of Covid-19 among 35 people out of 92 that they were in contact with each other at a church gathering on March 10th; three people later died. Refusal to implement pandemic - mitigation measures, in particular prohibiting public gatherings, at the very beginning led to increases in cases. Were the guidelines released and discussed earlier, the cases due to people gathering could have been preventable.



Graph 1.7-day incidence rate progression focusing on the period from the end of May to the beginning of October (Source: own analysis with data on the number of new cases from www.cdc.gov)

Even if it was far behind the magnitude of some other US states at the beginning, using NPI measures earlier on could have helped keep the virus at bay.
Even though individual states, like Arkansas, are clearly not independent, they can serve as small natural experiments in how different government responses can impact the course of an epidemic. For these reasons focusing on NPIs in individual states that implemented face coverings as one of the strategies of NPIs as well as on a hypothesis derived from literature we would like to test whether or not requiring wearing a mask has an impact on the seven day incidence rate in Arkansas compared to the control group states.

METHOD AND EMPIRICAL STRATEGY

For our analysis, we followed two different methods: Difference-in-Differences (DiD) design and the Synthetic Control Group (SCG) method.

DATA

During the pandemic, measurements such as case fatality rates, R-value, 7-day incidence rate, contagion index are important in understanding and predicting the course of the epidemic.

We begin our analysis by examining the dataset "United States COVID-19 Cases and Deaths by State over Time"[2] for the period January 21st 2020 to August 12th 2021 from the Centers for Disease Control and Prevention CDC website. This dataset records the aggregate counts of Covid-19 cases and deaths by states, territories and jurisdictions.
The CSV file contains the information we need for our analysis: number of new cases ordered by state and date.

In order to create the synthetic control group, data about the demography, social characteristics and health conditions are needed.
The American Community Survey (ACS) produces population, demographic and housing unit estimates, which are then presented by the Census Bureau's Population Estimates Program. We

[2] Available from:
https://data.cdc.gov/Case-Surveillance/United-States-COVID-19-Cases-and-Deaths-by-State-o/9mfq-cb36

download from the Table ID DP05 the 2019 estimates about male and female proportion, percentage of people aged under 18, over 18, over 65, over 85 and races. Also the population density estimates for each state are taken from the Census Bureau's Population Estimates Program[3].

The data about the health conditions are downloaded from the BRFSS Behavioral Risk Factor Surveillance System website[4]. The BRFSS is a system of health-related telephone surveys that collect data from the residents about health conditions and use of preventive service.
We are interested in the prevalence of diagnosed diabetes among adults aged over 18 years and the percentage of obese adults aged over 18 years.

To differentiate the State Government political party affiliation, we add to the dataset a binary variable which takes on the value of 1 if the State Governor's Party is Democratic and the value of 0 if it is Republican[5].

The outcome variable of this research is the 7-day incidence rate which is the average number of new infections from Covid-19 within the last seven days per 100,000 inhabitants, according to a positive PCT and antigen tests.
We downloaded the data about the new cases per 100,000 inhabitants from the Centers for Disease Control and Prevention CDC website and we created a new variable "cases_ma7" for the 7-day incidence rate.

DIFFERENCE IN DIFFERENCES METHOD

The Difference-in-Differences specification is a cross-sectional comparison statistical technique with before and after treatment. The specification uses longitudinal data from treatment and control groups in order to obtain an appropriate counterfactual to estimate the causal effect of a specific treatment (Angrist et al., 2009)
On the 20th of July 2020, the governor of Arkansas issued a state-wide face-covering mandate with the Executive Order 20-43: which would represent the treatment of interest in our research.

[3] https://www.census.gov/library/visualizations/interactive/2020-population-and-housing-state-data.html
[4] Data taken from database available from: https://www.cdc.gov/brfss/data_tools.htm
[5] From Ballotpedia website "Gubernatorial and legislative party control of state government", available from: https://ballotpedia.org/Gubernatorial_and_legislative_party_control_of_state_government

The model studied the effect of wearing masks on a 7 - days moving average number of new cases per 100,000 inhabitants (7 - day incidence rate) taking the state of Arkansas as a treatment group.

The aim is to compare the changes in outcome (7 - day incidence rate) over time between our treatment group (Arkansas) and the control group.

In order to find the most suitable control state to use in our Difference-in-Differences specification, we refer to the "Daily Kos Elections' State Similarity Index" published by the famous blog Daily Kos at the beginning of 2020.

David Jarman, the author of the article, settled a cross-section of 28 variables, including individual and social characteristics, political structure and adopted policies. Adding up all the differences across all categories, he obtained a list of less divergent states.

According to this analysis, the most similar states to Arkansas are Tennessee, Oklahoma and Missouri.

None of these three states has executed a state-wide face-covering mandate. Given this, they may represent a good control group.

The identification assumption of the DiD approach is that the group-specific trends in the outcome of interest would be identical in the absence of treatment.

Indeed, under the common trends assumption, the 7-day incidence rate of the control group reflects how the value would have changed in Arkansas if it didn't have implemented the face-covering mandate (Goodman-Bacon and Marcus, 2020).

We are proceeding analyzing the regression model. The dependent variable is the 7 - day incidence rate (every 100,000 inhabitants), *post* is the dummy variable for post period which takes on the value of 1 if post period: in this case, it represents the day of intervention (July 20, 2020). *tgroup* is the dummy variable which takes on the value of 1 for state in the treatment group. *DiD* represents the interaction term between the post period dummy and the treatment group dummy and its coefficient is the treatment effect.

$$y = \beta_0 + \alpha_0\, post + \beta_1\, tgroup + \delta\, DiD + u$$

We have proceeded running three different specifications, one for each state listed above.

	Missouri	Oklahoma	Tennessee
did	-29.667***	-3.647	-1.309
	(-3.607)	(-0.386)	(-0.122)
post	81.748***	55.728***	53.390***
	(14.055)	(8.341)	(7.059)
tgroup	47.667***	25.483***	-5.133
	(8.162)	(3.799)	(-0.676)
_cons	54.400***	76.583***	107.200***
	(13.173)	(16.144)	(19.962)
R-squared	0.589	0.395	0.293
N. obs	242	242	242

t statistics in parentheses
* p<0.05, ** p<0.01, *** p<0.001

Table 1. Estimated results output from Stata using a different control state for each specification

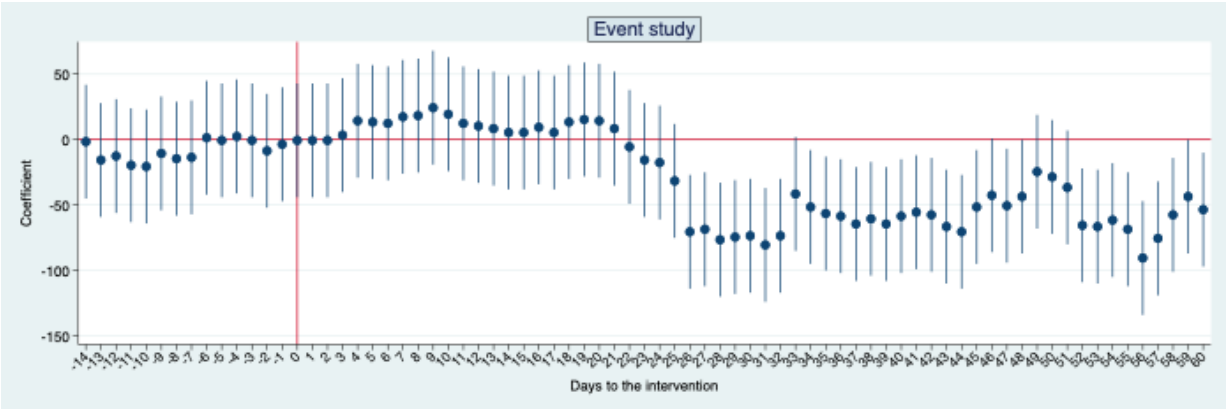
As result, the control group most suitable to our research question is Missouri, since the coefficients are highly significant.

In the absence of the intervention, the control group shows what would have happened in the treatment group if there was no treatment.

According to Table 1, the results of the Difference-in-Differences estimation indicate that face mask requirement (our intervention) has led to a significant variation in the 7 days incidence rate, with a decrease of almost 30 new cases in the 7 days incidence rate every 100,000 inhabitants.

We then need to focus on the event study using Missouri as a control group.

In order to get a more precise look at our event study, we are plotting the pre-treatment and post-treatment coefficients.



Graph 2. Event study coefficients from DiD specification (Source: own analysis with data on number of new cases from www.cdc.gov)

The Graph 2 figured above presents the pre-treatment "event study" and the post-treatment "event study" estimates of the coefficients.
The pre-treatment period is from July 6th 2020 and July 19th 2020, where the upper bound is given by the intervention day (July 20th 2020).
As noticeable, there is a decrease of new cases 10 days before the day of intervention: this might be because when the Governor of Arkansas announced policy ahead of time, the behavior of individuals may have changed as a response (Goodman-Bacon and Marcus, 2020).
The coefficient of the intervention day is nearly on the zero line itself, but the standard error is quite large. This means that the model does not estimate in a very precise way the difference, which should be zero, between the treatment and control group prior to the intervention.
Right after the intervention day, an increase in new cases has been recorded for the next first ten days; this bias might have been influenced by the general increase in new cases that were occurring in Arkansas in that period.

It is important to remember that every Covid-19 intervention needs some time before showing the effect on new cases. The virus has an incubation period which means that reported infections lag true infections by several days (Goodman-Bacon and Marcus, 2020). Indeed, previous studies have reported a median incubation period of 5.1 days, and symptoms may develop within 11.5 days in 97.5 percent of the cases (Lauer et al, 2020).
Twenty days after the July 20th 2020, the 7-day incidence index started to decrease.
This may represent the decreasing effect of mask requirement on new cases.

It is important to analyze what was happening in Missouri in that period of time and if Missouri is a good control group.
The Governor of Missouri signed an executive "stay-at-home" order on the 6th of April 2020, which was extended until the 3rd of May 2020. From the 4th of May, the state programmed an orderly reopening of businesses and community operations and resumed dining services.
So, no similar interventions were in place during our period of interest, thus this state fulfills the common trend assumption.

SYNTHETIC CONTROL METHOD

The analysis proceeds using the Synthetic Control Method instead to analyze the effect of face-covering on the spread of Covid-19 in Arkansas.

According to Abadie in his recent paper "Using Synthetic Controls: Feasibility, Data Requirements, and Methodological Aspects", the Synthetic Control Method provides a more appropriate control group when the treatment group is represented by small aggregate entities, like in this case.

The Synthetic Control Group is a control group that consists of "non-natural" units used to create counterfactuals for the treatment group, where these counterfactuals are constructed using a weighted average of the natural control units.

These weights are non-integer numbers, which sum up to 1 and are chosen by default by Stata in order to minimize some distance measures between treated and synthetic controls (Abadie et al., 2010).

The software "Stata" uses the command synth as a module to implement SCM for Comparative Case Studies (Stata, 2021). This command will plot the development of the outcome variable for Arkansas, as a treatment state, and its synthetic counterpart.

The idea behind the Comparative Case Studies is that the effect of treatment "can be inferred by comparing the evolution of the outcome variables of interest between the unit exposed to the treatment and a group of units that are similar to the exposed unit but were not affected by the treatment" (Abadie, 2021)

Indeed, with this method, the synthetic control units are intended to represent the counterfactuals of our case of interest.

In order to use the SCG method there are several requirements that need to be satisfied: there have to be multiple control units, several pretreatment observations and a single treatment time (Cunningham, 2021).

What is essential is to focus on the selection of the covariates that will be used for any matching.

With the choice of covariates, the weights are optimally chosen to minimize the distance in value between the treated unit and its synthetic control. (Cunningham, 2021)

The predictors are the median age, percentage of people aged over 65, percentage of people aged over 85, diabetes rate, obesity rate, population density, state governor's party affiliation and the outcome (7-day incidence rate) for each of the previous 15 days.

The synthetic Arkansas is constructed using these predictor variables, as shown in Table 2.

	Treated	Synthetic
medianage	38.8	39.384
ageover65	.174	.175474
ageover85	.019	.018875
diabetes_rate	.136	.121155
obesity_rate	.374	.353637
pop_density	57.9	121.2661
government	0	.554
cases_ma7	161	162.217
l1cases_ma7	155	155.265
l2cases_ma7	164	166.904
l3cases_ma7	168	168.212
l4cases_ma7	168	167.061
l5cases_ma7	173	165.335
l6cases_ma7	156	156.668
l7cases_ma7	152	154.172
l8cases_ma7	155	152.429
l9cases_ma7	139	140.61
l10cases_ma7	132	133.385
l11cases_ma7	134	131.242
l12cases_ma7	124	128.856
l13cases_ma7	132	131.741
l14cases_ma7	132	129.411

Table 2. Predictor balance Synthetic Control Method, Specification (1)

The potential control units are represented by the other 49 states of the United States of America.

The Stata output gives a detailed analysis about the contribution of each potential control units in the donor pool to the synthetic control for Arkansas.

Co_No	Unit_weight
LA Louisiana	.173
MN Minnesota	.095
NC North Carolina	.162
NE Nebraska	.124
SC South Carolina	.258
WV West Virginia	.187

Table 3. Unit weights Synthetic Control Method, Specification (1)

The states that are not included in the list do not contribute to the synthetic group.

Abadie highlights that to provide a realistic control, it is important that the units included in the donor pool have been chosen judiciously (Abadie, 2021): we have to restrict the donor pool to units with characteristics that are similar to the affected unit, otherwise the result would be biased.

Analyzing the countries that are part of the donor pool, we have to check if similar interventions were issued during that period.

In Louisiana, Minnesota, North Carolina and in West Virginia a mask requirement was issued in that period of time.

Four out of six potential control units experienced the same intervention during the same period of the year. This represents a violation of the common trend assumption, since the treatment also affects the synthetic control group.

We proceed by eliminating the potential control units that violate the common trend assumption. Rerunning the synth command, we obtain a new synthetic control group.

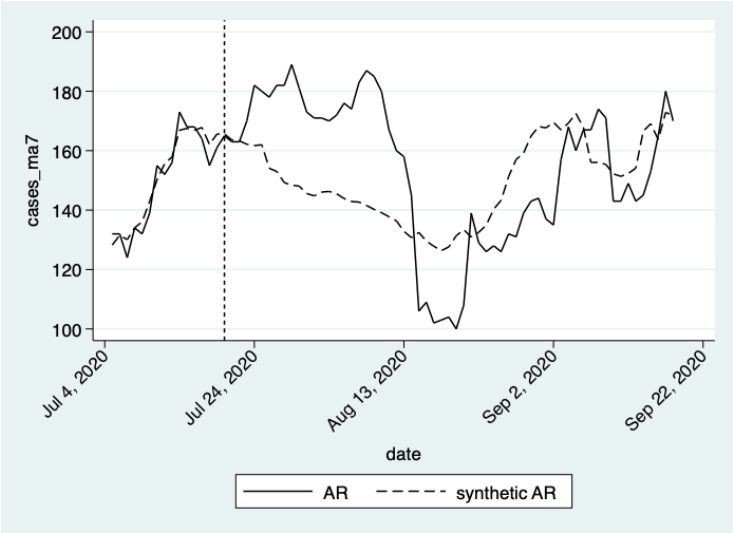
	Treated	Synthetic
medianage	38.8	38.7654
ageover65	.174	.173011
ageover85	.019	.019628
diabetes_rate	.136	.117197
obesity_rate	.374	.350673
pop_density	57.9	108.6966
government	0	.119
cases_ma7	161	165.581
l1cases_ma7	155	162.098
l2cases_ma7	164	167.829
l3cases_ma7	168	166.853
l4cases_ma7	168	167.514
l5cases_ma7	173	166.895
l6cases_ma7	156	157.927
l7cases_ma7	152	155.608
l8cases_ma7	155	150.237
l9cases_ma7	139	142.97
l10cases_ma7	132	136.082
l11cases_ma7	134	134.05
l12cases_ma7	124	130.068
l13cases_ma7	132	131.508
l14cases_ma7	132	128.241

Table 4. Predictor balance Synthetic Control Method, Specification (2)

Co_No	Unit_weight
IA (Iowa)	.079
MO (Missouri)	.414
ND (North Dakota)	.119
SC (South Carolina)	.292
TN (Tennessee)	.096

Table 5. Unit weights Synthetic Control Method, Specification (2)

In this case, the synthetic control group fulfills the common trend assumption since no similar treatment was issued in that period in any of those countries.

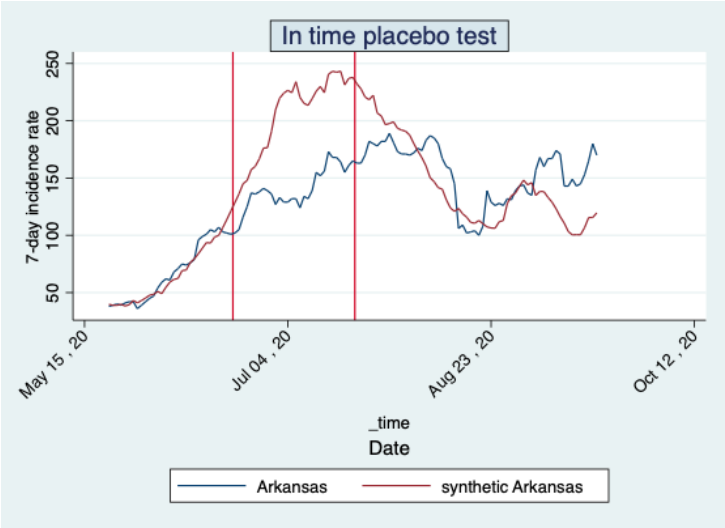


Graph 3. Synthetic Control Method, Specification (2)

Looking at Graph 3, the pre-treatment trend is very similar for both treatment and control groups. Few days after the intervention day, the expected decrease in the 7-day incidence rate does not occur. Instead, the result is different from what we were expecting.
The introduction of a mask requirement has not decreased the new cases in Arkansas, relative to the synthetic control group. A huge drop was registered more than 3 weeks later the intervention day, which suggests that the decrease may not be caused directly by the treatment itself.

In Comparative Cases Studies, the use of statistical inference is quite difficult. According to Abadie in "Comparative Politics and the Synthetic Control Method", this is mainly because of the small sample nature of the data and lack of randomization.
The researchers often conduct some falsification exercises, which are called "placebo studies". There are two different types of robustness tests: in time placebo test and in space placebo test. In the *in time placebo test*, the researcher pretends that the intervention took place at an earlier point in time; while in the *in space placebo test*, the researcher pretends that the intervention took place in a control unit at the same time.

The first robustness test to compute is the in time placebo test.
It can be tested whether the Synthetic Control method produces large estimated effects if the treatment was issued 30 days before. If the effect is similar or larger with respect to the original simulation (treatment was implemented on July 20th, 2020), the confidence that the effect estimated for the 20th July is attributable to the treatment itself would significantly shrink.

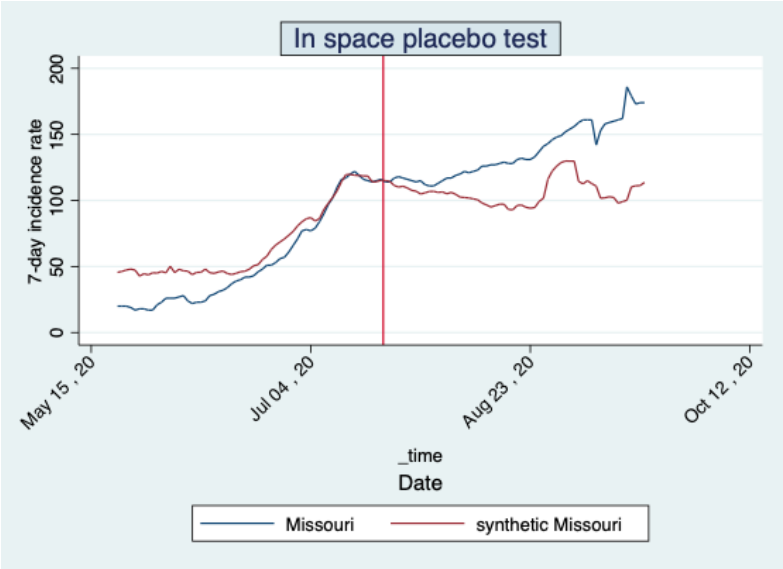


Graph 4. In time placebo test: the intervention artificially takes place one month earlier

The two lines from Graph 4, should be parallel during the period between the placebo intervention day and the actual intervention day.
In this case, the in time placebo study would suggest that synthetic controls do not provide good predictions of the trajectory of the outcome in Arkansas in periods when the treatment was not in place, since Arkansas and its synthetic control deviate even before the actual intervention.

This method shows one of the biggest problems in empirical research on Covid-19 interventions, since interventions are often taken based on the trend of new cases: when the 7-day incidence rate is increasing, governments decide to implement some new measures in order to contain the spread of the infections.

Another way to conduct some placebo tests is to reassign the intervention not in time, but to a member of the donor pool.
In the in space placebo test, instead of using Arkansas as a treated unit, we substitute it with Missouri.
The Graph 5 shows that the two lines are parallel until 15 days after the intervention day. Then, the outcome in Missouri starts increasing while in its synthetic counterfactual it is decreasing.
The confidence that the face covering intervention has an effect on the outcome is rejected if similar estimates arise when the intervention artificially took place in a control unit.



Graph 5. In space placebo test: the intervention artificially takes place in the control state Missouri

CONCLUSION

The Difference-in-Difference design has proven a significant decrease in 7-day incidence rate and consequently in new daily cases, after the issue of an Executive Order for a state-wide mask requirement.

However, using the Synthetic Control Method, the graph shows an unexpected increase in the new cases and a later decrease twenty days after the intervention date, but after running some placebo regressions, we cannot declare that the change in the 7-day incidence rate is due to the treatment itself.

Despite the findings of our research does not provide a strong evidence of the decrease in contagion thanks to the mask requirement, the World Health Organization continues to highlight how masks should be "part of a comprehensive strategy of measures to suppress transmission and save lives" (WHO, 2020). Indeed, the use of mask alone without any other containment measure is not sufficient to provide a good level of protection. Masks are particular effective in limiting the spread of infections in combination with other preventive measures such as physical distancing, high ventilation of the spaces and personal hygiene.

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TASK BREAKDOWN

After having decided together the topic to develop for our term paper,
Lisa Fadelli has collected all data needed to implement our research, created the do-file with the coding to replicate the paper, prepared the replication package and the README file. She has edited the "Method and Empirical Strategy" paragraph, the abstract and the conclusion.
Sergei Sviridov has done researches on the literature about our topic. He has written the paragraphs "Introduction", "Literature review", "Covid-19 situation in Arkansas". However, the entire project was under the supervision of both of us.
Together, we have prepared the paper template and the structure, and we have completed the "References" part.

DECLARATION AGAINST PLAGIARISM

Herewith, we confirm that we have written the term paper independently and without help from another party. All text passages taken word-for-word or their meaning from other publications or the World Wide Web are identified as such. The provided replication package produces all relevant Table and Figures in the text.

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31st August 2021

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